

Tide retrospective:

Bounded-prior quartic with continuous test function

Daniel Murfet (with Claude Opus 4.7)

6 May 2026

Abstract

A single Tide-loop excursion in the `laplace` seabed, advancing the grammar paper precursor sequence by one step: where Tide 6 added the bounded prior to the pure-quartic partition function, this step adds a Lipschitz test function φ inside the integrand. The headline¹ proves $|\int_{[-a,a]} \varphi(w) e^{-tw^4/24} dw - \varphi(0) Z_a(t)| \leq K \cdot \frac{1}{2} \sqrt{24\pi/t}$ for K -Lipschitz φ . The new file adds 211 lines, zero `sorry/axiom/native.decide`; clean Lean diagnostics on first pass after two small `Pi-vs-lambda` fixes. The proof structure follows GPT-5.5 Pro’s deliberation-stage tactical advice (“prove the unnormalised estimate first, derive the normalised version as a corollary”), which sidesteps a denominator-conditioning subtlety that trapped my initial *normalised* draft. Run executed in a dedicated `git worktree` — the discipline now mandated by the Tide skill — with no concurrent-agent races.

1 Setting

This Tide step is the seventh excursion outward from the `laplace` seabed and the second toward the grammar paper. The earlier `tide/grammar-precursor-bounded-prior` excursion (2 May) established the bounded-prior quartic partition function with an exponentially small remainder.² That excursion’s Follow-up section explicitly named the present candidate as its natural successor: continuous (or Lipschitz) test functions φ against the same bounded-prior quartic Gibbs measure.

The Step 2 deliberation — which started by drafting three candidate scopings — landed on a sharpening that wasn’t on Claude’s initial draft: prove the *unnormalised* Lipschitz estimate $|\int \varphi g - \varphi(0) \int g| \leq K \frac{1}{2} \sqrt{24\pi/t}$ first, then derive the normalised $|\langle \varphi \rangle_{t,a} - \varphi(0)| = O(t^{-1/4})$ as a corollary. This is GPT’s contribution and was the right call (see §*Roadblocks* below for why). The full deliberation, with the GPT consult preserved verbatim, lives in the primer project’s tide-log.³

The branch is chained on top of the `tide/separable-potential` tip, itself one of the two unmerged tides at the start of this excursion (the other was `tide/cross-susc-deriv-harmonic`, on a parallel and independent piece of laplace–threepoint cross-work). The linear-chain discipline — each new Tide branched off the previous one’s tip rather than off `main` — keeps the squash-merge story straightforward.

¹`quartic_lipschitz_unnormalised_bounded_prior`.

²Headline: $|\int_{[-a,a]} e^{-tw^4/24} dw - \frac{1}{2} (24/t)^{1/4} \Gamma(1/4)| \leq \frac{24}{ta^3} e^{-ta^4/24}$, in `Laplace.OneD.QuarticBoundedPrior`.

³`projects/primer/tide-log/2026-05-06-tide-bounded-prior-continuous-test.md`.

2 The thing formalised

For the centred quartic potential $L(w) = w^4/24$, any $a, t > 0$, $K \geq 0$, and any continuous $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ satisfying the Lipschitz-near-zero condition

$$|\varphi(w) - \varphi(0)| \leq K \cdot |w| \quad \text{for all } w \in [-a, a],$$

the bounded-prior unnormalised numerator differs from $\varphi(0)Z_a(t)$ by at most a closed-form constant times $t^{-1/2}$:

$$\left| \int_{[-a, a]} \varphi(w) e^{-tw^4/24} dw - \varphi(0) \int_{[-a, a]} e^{-tw^4/24} dw \right| \leq K \cdot \frac{1}{2} \cdot \sqrt{24\pi/t}.$$

The Lean signature:

```
theorem quartic_lipschitz_unnormalised_bounded_prior
  {t a K : ℝ} (ht : 0 < t) (ha : 0 < a) (hK : 0 ≤ K)
  {φ : ℝ → ℝ} (hφ : Continuous φ)
  (hLip : w ∈ Icc (-a) a, |φ w - φ 0| ≤ K * |w|) :
  |(w in Icc (-a) a, φ w * exp (-(t * w ^ 4 / 24))) -
   φ 0 * (w in Icc (-a) a, exp (-(t * w ^ 4 / 24)))| ≤
  K * (1/2) * Real.sqrt (24 * Real.pi / t)
```

located in `Laplace.OneD`.

The grammar paper’s $Z_n[\varphi]$ machinery wants three ingredients on top of the existing seabed: (i) bounded prior π , (ii) test function φ inside the integrand, (iii) explicit asymptotic in n with a remainder. Tide 6 added (i); the present tide adds (ii). What remains is the explicit $o(t^{-1/4})$ packaging of (iii), now reachable as a follow-up that combines this tide’s bound with the existing `quartic_partition_bounded_prior` (an exponentially small correction to $Z_a(t)$).

The unnormalised rate is $t^{-1/2}$ in this statement; the normalised $|\langle \varphi \rangle_{t,a} - \varphi(0)|$ then gets the “substitution-driven” rate $t^{-1/4}$ once one divides by $Z_a(t) \sim Z_{\text{full}}(t) = O(t^{-1/4})$. The closed-form rate constant for the normalised version is $C = 24^{1/4} \sqrt{\pi} / \Gamma(1/4) \approx 1.082$, but that is a corollary, not the headline.

Numerical sanity check (run before formalisation): `scipy.integrate.quad` verified all three deliberation-stage candidates’ closed forms to machine precision across five values of t ranging from 10 to 10^5 , with several test functions.⁴ The β -rate constant C matched the predicted value of 1.082048 to machine precision for $\varphi = |w|$ at $t \geq 1000$.

3 Proof strategy

The unnormalised statement reduces, via the trivial identity

$$\int \varphi g - \varphi(0) \int g = \int (\varphi - \varphi(0))g,$$

to bounding $|\int (\varphi - \varphi(0))g|$. Pointwise on $[-a, a]$, $|(\varphi - \varphi(0))g| \leq K \cdot |w| \cdot g$ by the Lipschitz hypothesis (the Boltzmann factor $g(w) = e^{-tw^4/24}$ is positive, so its absolute value is itself). The

⁴ $\varphi \in \{1, w, w^2, |w|, \cos w, \cos w + \frac{1}{2}\}$.

triangle-for-integrals lemma $|\int f| \leq \int |f|$ then takes the absolute value inside, and the pointwise bound followed by monotonicity of the integral (`setIntegral_mono_on`) reduces the whole thing to bounding $\int_{[-a,a]} |w| g(w) dw$.

The bounded-prior first absolute moment is bounded by its full-line counterpart by monotonicity over a subset (the integrand is non-negative). The full-line first absolute moment has a clean closed form:

$$\int_{\mathbb{R}} |w| e^{-tw^4/24} dw = \frac{1}{2} \sqrt{24\pi/t}.$$

This is the substantive new computation in this tide. The proof goes through the half-line case $\int_{(0,\infty)} w e^{-tw^4/24} dw = \frac{1}{4} \sqrt{24\pi/t}$, which falls directly out of Mathlib’s master gamma-integral lemma⁵ at $p = 4$, $q = 1$, $b = t/24$, with the closed-form value $(t/24)^{-1/2} \frac{1}{4} \Gamma(1/2)$, which simplifies to $\frac{1}{4} \sqrt{24\pi/t}$ (since $\Gamma(1/2) = \sqrt{\pi}$ and $(t/24)^{-1/2} = \sqrt{24/t}$). The full-line case is twice the half-line case by parity of the integrand, via Mathlib’s `integral_comp_abs`—exactly the same routing established by the Quartic and Sextic moment files.

So the chain is:

1. Half-line first moment, closed form (master lemma).
2. Full-line first absolute moment, double of (1) by parity.
3. Bounded-prior first absolute moment $\leq$ full-line, by set monotonicity.
4. Apply Lipschitz pointwise, triangle-inequality, set monotonicity, and (3).

Each link is a single Mathlib invocation modulo bookkeeping; the file is 211 lines because the bookkeeping (rpow-vs-npow casts, Pi.sub-vs-lambda mismatches, integrability witnesses) is unavoidable at present.

4 Roadblocks and resolutions

The deliberation-stage pivot from normalised to unnormalised. Claude’s initial Step 2 draft proposed the normalised statement $|\langle \varphi \rangle_{t,a} - \varphi(0)| \leq KCt^{-1/4}$ as the target. GPT-5.5 Pro pointed out that proving this directly trips on a subtle conditioning identity:

GPT-5.5 Pro: The truncated Gibbs law on $[-a, a]$ is the full-line quartic law conditioned on $|W| \leq a$, so its $|W|$ -expectation is $\leq$ the full-line $|W|$ -expectation. That full-line expectation is exactly $Ct^{-1/4}$.

That paragraph reads cleanly, but *it is not quite right*: for a non-negative random variable Y and an event A , the conditional expectation $E[Y \mid A] = E[Y \cdot \mathbf{1}_A]/P(A)$, which is bounded *above* by $E[Y]/P(A)$, not by $E[Y]$. For our problem $P(|W| \leq a) \rightarrow 1$ exponentially as $t \rightarrow \infty$, so $1/P(|W| \leq a) = 1 + (\text{exp-small})$ and the normalised bound recovers an exp-small correction GPT’s first-pass argument suppresses.

The unnormalised statement, by contrast, dodges all conditioning: the inequality

$$\int_{[-a,a]} |w| g(w) dw \leq \int_{\mathbb{R}} |w| g(w) dw$$

⁵`integral_rpow_mul_exp_neg_mul_rpow`.

follows from set monotonicity of the integral with no probability arithmetic at all. GPT’s broader recommendation — “prove the unnormalised version first, then derive the normalised one as a corollary” — is exactly right, and it’s the right architectural choice even if the specific justification needs a small correction. The unnormalised version landed cleanly here; the normalised corollary now becomes a separate, smaller follow-up tide that can carry the partition ratio analysis explicitly.

Two Pi-vs-lambda mismatches. Both familiar from the project’s `CLAUDE.md` cookbook. The first appeared in the integrability of $|w| \cdot g(w)$: `Integrable.abs` returns the integrability of $|w \cdot g(w)|$, but the headline wants $|w| \cdot g(w)$, and converting between them needs $|w \cdot g| = |w| \cdot |g| = |w| \cdot g$ (since $g > 0$). The fix is the standard `abs_mul` $\rightarrow$ `abs_of_pos (Real.exp_pos _)` rewrite inside a `funext`. The second appeared in subtracting two integrable functions on $[-a, a]$: `IntegrableOn.sub` returns `IntegrableOn (f - g) ...` with $f - g$ as `Pi.sub`, but the headline expects the single-lambda `fun w => (φ w - φ 0) · g w`. The fix is to prove a `funext` equality and `rwa`; same recipe as the quartic-cov-affine and pure-quartic tides. Both fixes were one-shot.

The rpow-to-sqrt conversion. Going from $(t/24)^{-1/2}$ to $\sqrt{24/t}$ took a few attempts through different Mathlib idioms before landing on a four-step rewrite chain.⁶ This sequence is not in the project’s `CLAUDE.md` yet; should be added.

Worktree isolation worked as designed. The Tide skill now mandates running each Tide in its own `git worktree`. This run was the first to use that discipline from the start: the worktree at `/tmp/laplace-bounded-test/` was created before any work began, and no concurrent-agent races materialised. Compare with the earlier 2 May Tides where multiple concurrent agents racing on `Laplace.lean` cost an extra round-trip per Tide. The lesson is now load-bearing in the skill, not a footnote.

5 What was Mathlib, what was new

Mathlib provided. The master half-line gamma integral; the even-symmetric reduction to the half line; the closed-form $\Gamma(1/2) = \sqrt{\pi}$; the triangle inequality $|\int f| \leq \int |f|$; pointwise-bound monotonicity; set monotonicity (subset `Icc` into all of $\mathbb{R}$); the `rpow-to-sqrt` conversion lemmas; the integration-by-parts lemmas for linear combinations.⁷

The seabed provided. `quartic.integrable_pow` from `Laplace.OneD.Quartic` (for the full-line integrability of $w^n e^{-tw^4/24}$). The tide’s import is just `Laplace.OneD.QuarticBoundedPrior`; the partition function and `quartic.integrable_pow` chain transitively from there.

What was new in this tide. The half-line first moment, the full-line first absolute moment, the integrability witness, the bounded-prior comparison, and the headline unnormalised Lipschitz estimate.⁸ Five new theorems; the four sublemmas are general enough they could be promoted to `lean-common` on the next bounded-prior tide that needs them.

⁶`Real.rpow_neg` $\rightarrow$ `Real.sqrt_eq_rpow` $\rightarrow$ `Real.sqrt_inv` $\rightarrow$ `congr 1`; `field_simp`.

⁷Specifically: `integral_rpow_mul_exp_neg_mul_rpow`, `integral_comp_abs`, `Real.Gamma_one_half_eq`, `MeasureTheory.abs_integral_le_integral_abs`, `MeasureTheory.setIntegral_mono_on`, `MeasureTheory.setIntegral_le_integral`, `Real.rpow_neg`, `Real.sqrt_eq_rpow`, `Real.sqrt_inv`.

⁸Lean names: `quartic.integral_w_exp_Ioi`, `quartic.integral_abs_w_exp_full`, `quartic.integrable_abs_w`, `quartic.integral_abs_w_bounded_prior_le`, `quartic.lipschitz_unnormalised_bounded_prior`.

6 Lessons

- *Unnormalised before normalised.* The Step 2 deliberation taught the right architectural lesson for grammar-paper-style asymptotics: when the headline statement is a normalised ratio $\langle \varphi \rangle_{t,a} \rightarrow \varphi(0)$, the cleanest formalisation path goes through the unnormalised numerator difference first. The normalisation—dividing by the partition function—brings in denominator gymnastics (lower bound on Z_a , ratio of two limits) that are easier to handle once the unnormalised piece exists as a stand-alone theorem. Worth lifting to the Tide skill or to `lean-formalisation`’s playbook.
- *Trust GPT’s strategic recommendations even when the justification has a small flaw.* GPT’s “the truncated Gibbs is conditioned on $|W| \leq a$, so its $|W|$ -expectation is $\leq$ the full-line one” was not quite right (it’s bounded by $E[|W|]/P(|W| \leq a)$, with a $1/P$ factor that is benign but not invisible). But the broader architectural recommendation—prove unnormalised first—was completely right, and the right mental move was to take the architectural advice and substitute the cleaner *set-monotonicity* justification for the sketch GPT gave.
- *The rpow-to-sqrt cast chain.* The four-step rewrite⁹ recurs in any Tide that converts a half-line $\Gamma(1/2)$ result to its sqrt presentation. Worth a one-line entry in `laplace`’s `CLAUDE.md` so the next session doesn’t re-derive it.
- *Worktree isolation is now load-bearing.* The Tide skill’s preflight rule (every Tide in its own `git worktree`) was followed from the start and the run was free of branch-toggle races. This is the first run that hit zero concurrent-agent friction. The discipline is paying for itself.

7 Follow-ups

- *Normalised version:* $|\langle \varphi \rangle_{t,a} - \varphi(0)| \leq KCt^{-1/4} + (\text{exp-small})$. Direct corollary combining the present unnormalised estimate with the existing `quartic_partition_bounded_prior`: lower-bound $Z_a(t)$ by $(1/2)(24/t)^{1/4}\Gamma(1/4) - \frac{24}{ta^3}e^{-ta^4/24}$, divide. The explicit constant $C = 24^{1/4}\sqrt{\pi}/\Gamma(1/4)$ is the rate constant; numerically $C \approx 1.082048$. A quick follow-up tide (~ 60 – 100 lines).
- *IsLittle0 packaging.* For continuous φ (weaker than Lipschitz), the limit $\langle \varphi \rangle_{t,a} \rightarrow \varphi(0)$ holds qualitatively—the rate $t^{-1/4}$ is special to Lipschitz φ —and Mathlib’s asymptotic-equivalence API¹⁰ would be the form needed for the grammar paper’s $Z_n[\varphi] \sim C_\varphi n^{-\lambda}(\log n)^{m-1}$. This is a separate, larger Tide.
- *Bounded-prior monomial generalisation* $L = w^{2k}/(2k)!$. The full chain transfers verbatim to the sextic ($k = 3$), octic ($k = 4$), and so on. Pairs naturally with the morning survey’s A1 candidate (“generic even-monomial 1D template”), which is now strict-improvement territory: with three k -instances completed (quartic, sextic, and now the bounded-prior version of the quartic), the case for a unifying parameterised file is overwhelming.
- *Promote the splitting/union and abs-of-product idioms.* Several small lemmas in this file (`abs_mul + abs_of_pos` for an exp-positive Boltzmann factor; the $|w \cdot g| = |w| \cdot |g|$ rewrite chain; the unnormalised-difference $= \int (\varphi - \varphi(0))g$ identity) recur on every Tide that involves

⁹Same sequence as above: `Real.rpow_neg` $\rightarrow$ `Real.sqrt_eq_rpow` $\rightarrow$ `Real.sqrt_inv` $\rightarrow$ `congr 1`; `field_simp`.

¹⁰`Asymptotics.IsLittle0`.

a Lipschitz observable against a Gibbs measure. Worth promoting to `lean-common` on the next use.

- *Continuous (not Lipschitz) test function via modulus-of-continuity.* GPT’s missed candidate (i): replace the Lipschitz-rate hypothesis with a modulus-of-continuity statement. The qualitative limit (α) becomes a clean ε/δ argument; smoothing parameter δ chosen via the tail bound. Cleaner formulation for “continuous test functions” than the Lipschitz cut, but larger formalisation.

Acknowledgements

GPT-5.5 Pro contributed the deliberation-stage advisory at Step 2, including the architectural recommendation to prove the unnormalised estimate first (the recommendation that landed) and the sharpened-but-slightly-imprecise justification for it (the conditioning argument that needed correcting on the way to the proof). The full consult and the tide log itself are preserved together in the primer project’s tide-log directory.¹¹

¹¹`projects/primer/tide-log/gpt55_tide.bounded_prior_continuous.test.v1.md` and the markdown tide log alongside it.